

An Axiomatic Model III: Axiomatic Derivation of the Structural Constants of Black Hole Thermodynamics - the 4 in $S = A/4$, the Evaporation Coefficient 5120, and the Imprint of Absent Time

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Abstract: Four axioms (orthogonality, atomicity, minimal cost, causality preservation) and one proposition define a discrete computational system. Because no free parameters enter the deduction, fitting is excluded; because the axioms close the search space, numerological tuning is blocked. The present paper is the third in the Axiomatic Model series and proceeds on the same axiom set as Paper I [5] and Paper II [7]. The complete axiomatic system comprises 15 axioms [6]; the present paper additionally uses its number registry (complete descriptive degrees of freedom), its recovery rule, and its juim-recording rule. The paper treats three structural constants of black hole thermodynamics. First, the coefficient 4 of the Bekenstein-Hawking entropy $S = A/(4\ell_p^2)$ is the structural-bracket domain count 4, while the 4 of the quasinormal-mode (QNM) asymptotic damping spacing $M\Delta\omega_I = 1/4$ is the cost-bracket ball value 4 ($= 1$ time + 3 space) - the two 4s are different quantities registered in different brackets, coinciding numerically only in three spatial dimensions (Theorem 1). A discrimination criterion follows: the ball value varies as $1 + d$ with spatial dimension d , whereas the entropy 4 is dimension-independent, so higher-dimensional holography separates the two readings. Second, in the bit reading (observer, superposition, time, space) $= (0, 1, 0, 1)$ of Paper II Theorem 1, time $= 0$, so the ball value 4 cannot be procured on that slice; hence obtaining the entropy coefficient from the Wheeler-DeWitt equation forces a discrete scale to be injected from outside the equation (Theorem 2) - the methodological constraint that Vaz's canonical quantization of the Schwarzschild black hole [8] had to introduce a discrete mass spectrum as an external assumption is a post-hoc concordance of this theorem. Third, the temperature-lifetime identity $T_H^3 \tau_{BH} = (10/\pi^2) T_P^3 t_P$ holds, with the mass cancelling exactly as $M^{-3} \times M^3$ (an identity by definition), and the evaporation coefficient decomposes as $5120 = 10 \times 2^9$ - where $10 = \dim SO(5)$ is the dimension of the maximal compact subgroup $SO(5)$ of the very signature $(5, 2)$ from which Paper I obtained the fine-structure constant, and the 9 of 2^9 is the structural complete descriptive degrees of freedom $9 = 7 + 2$ (Theorem 3). The event-horizon radius is reproduced as $r_s = 2GM/c^2$ by stacking N contraction units of $2\ell_p$, one per juim from its two cost-bearing stages (an identity by definition, Theorem 4), wrapped by the picture of a pairwise-cumulative-cost saturation surface (class III). As auxiliary results we organize, taking the results of Paper II as input, the Barbero-Immirzi $\gamma = \sqrt{3} \ln 3/8$ (a definite deviation of 0.136% from the transcendental-equation root of the literature), the LQG area gap, and the QNM real part, and we show that the well-known mismatch between the LQG area quantum and Hod's area quantum reduces, after cancellation of $\ln 3$, to the single geometric factor $3\pi/8$. The derivation yields 4 unique predictions with explicit falsification conditions (Table 6). Since Paper I obtained the electromagnetic coupling constant from $(5, 2)$ and Paper II obtained the metric of gravitational quantization from the same $(5, 2)$, the present result is a third physical domain of the same sign structure - that the electromagnetic coupling constant, the metric of gravitational quantization, and black hole thermodynamics descend from the same root is the structural conclusion of this series.

Keywords: black hole thermodynamics; Bekenstein-Hawking entropy; evaporation coefficient; quasinormal modes; Barbero-Immirzi parameter; Wheeler-DeWitt equation; Compare-And-Swap; computational primitives in physics

1. Introduction

Black hole thermodynamics carries three celebrated numbers. The first is the denominator 4 of the Bekenstein-Hawking entropy

$$S = \frac{k_B c^3 A}{4 G \hbar} = \frac{A}{4 \ell_P^2} \quad (1)$$

[1,2]. In standard physics this 4 is a derived quantity that falls out of the Euclidean path integral or of the horizon-temperature computation; no viewpoint in standard physics interrogates the number itself as an independent structural constant. The second is the coefficient 5120 of the Schwarzschild evaporation lifetime

$$\tau_{BH} = \frac{5120 \pi G^2 M^3}{\hbar c^4} \quad (2)$$

[2,3]. This number falls out of the integral combining the Stefan-Boltzmann law with the Schwarzschild metric (the textbook coefficient of the photon black-body approximation - see the qualification in Section 9); no literature treats the number itself as a structural constant. The third is the QNM asymptotic damping spacing

$$M \Delta \omega_I = \frac{1}{4} \quad (3)$$

established by Nollert's numerical discovery [16] and Motl's analytic proof [15]; this 4 sits at the center of the long-standing tension between the loop-quantum-gravity (LQG) area quantum and Hod's area quantum [17].

The present paper derives and interprets these three numbers from a single axiom set. On the same 4 axioms and 1 proposition used in Paper I [5] and Paper II [7], we additionally use the number registry (complete descriptive degrees of freedom), the recovery rule, and the juim-recording rule of the complete system [6]. The results are organized into four theorems.

- **Theorem 1 (bracket separation):** the 4 of Eq. (1) and the 4 of Eq. (3) are not the same number. The former is the domain count of the structural bracket, the latter the ball value of the cost bracket, and the two brackets are orthogonal by Axiom 1. They coincide numerically only in three spatial dimensions and separate in higher dimensions - this separation supplies a testable discrimination criterion.
- **Theorem 2 (imprint of absent time):** in the bit reading of Paper II Theorem 1, time = 0, so the ball value 4 cannot be procured inside the Wheeler-DeWitt slice. Obtaining the entropy coefficient from that equation therefore forces a discrete scale to be injected from outside. The methodological choice of the independent literature [8] is a post-hoc concordance of this theorem.
- **Theorem 3 (temperature-lifetime identity):** $T_H^3 \tau_{BH} = (10/\pi^2) T_P^3 t_P$, an identity in which the mass cancels exactly; the 10 of $5120 = 10 \times 2^9$ is $\dim SO(5)$, the group dimension of the signature (5,2) from which Paper I obtained the fine-structure constant.
- **Theorem 4 (contraction stacking):** stacking N contraction units of $2\ell_P$ - one per juim from its two cost-bearing stages - reproduces $r_s = 2GM/c^2$ as an identity. The picture that reads the impossibility of information escape as the blocking of the Read path rather than the divergence of the Read cost (the cumulative-cost picture, class III) wraps this identity.

Position of the paper. This is the third paper of the Axiomatic Model series. Paper I [5] showed that the irreversible/reversible cost classification forces the quadratic-form signature (5,2) (Paper I Theorem 1) and reached the fine-structure constant $\alpha = 1/137.036\,082$ (6×10^{-7} against experiment) through the bounded symmetric domain D_5 . Paper II [7] forced the bit reading (0,1,0,1) of the Wheeler-DeWitt equation uniquely from the same (5,2) (Paper II Theorem 1) and derived the component count 6 and the signature (5,1) of the DeWitt metric. The present paper treats a third physical domain of the same structure: black hole thermodynamics. The $\dim SO(5) = 10$ of Paper I reappears in the evaporation

coefficient (Theorem 3), and the bit reading of Paper II shows, for the first time, explanatory power beyond postdiction (Theorem 2).

Status classification. As in Paper I [5], the epistemic status of each result is classified in four tiers: *I. Axiom-derived* (formal proof or completed forward chain), *II. Axiom-determined* (the axioms force a structure and mathematics then fixes the value automatically), *III. Axiom-described* (derivation path and interpretation are in the text but an independent formal proof is deferred), *IV. Out of scope*. Each result carries its tier, and Section 10 tallies them.

Prior record. We state the recording dates of the mining cards used in this paper. The temperature-lifetime identity (D-32), the evaporation-coefficient decomposition (H-54), the domain reading of the entropy coefficient (H-71), the Schwarzschild radius (D-46), and the horizon cost boundary (D-49) were recorded during the first mining period of 21-26 March 2026 (the repository record of the decomposition $5120 = 10 \times 2^9$ is dated 24 March 2026) and were included in the publicly released edition of 26 March 2026 (v1.52, zenodo record 19222015) [6]. The Barbero-Immirzi card (D-159), the LQG area gap (D-160), the QNM synthesis (D-161), and the separation of the two 4s (H-979) were recorded on 18 August 2026 and included in the current edition of the complete system (v1.8, record 20301304) [6]; of these, the gamma, the area gap, and the QNM real part were already published as Section 7.1 of Paper II [7] (released 29 August 2026).

2. Axiomatic System

2.1. Relation to the Earlier Papers

The axioms of this paper are exactly the same set as Section 2 of Paper I [5] and Section 2 of Paper II [7]. The complete axiomatic system comprises 15 axioms [6]; Papers I and II produced their results on 4 of them (numbers 1, 2, 4, 15 of the complete system) plus 1 proposition. The present paper uses the same 4 axioms and 1 proposition as its base, and additionally uses three items of the complete system: the number registry (complete descriptive degrees of freedom, Axiom 9 of the complete system), the recovery rule (Axiom 6), and the juim-recording rule (Axiom 7). The additional items are listed exhaustively in Section 2.3. The summary below follows the rendering of Paper II Section 2.2; the full statements and propositions are in Paper I and in the complete system [6]. The full text of the 15 axioms and their propositions can be read online (<https://ubmscoin.github.io/banya/en/axiom.html>).

2.2. The 4 Explicit Axioms + 1 Proposition (Summary Rendering)

Axiom 1 (Banya equation - 4-axis orthogonal norm). *Every change δ of the universe is the norm of 4 mutually orthogonal axes:*

$$\delta^2 = (\text{time} + \text{space})^2 + (\text{observer} + \text{superposition})^2. \quad (4)$$

The first bracket is called the classical bracket (DATA), the second the quantum bracket (OPERATOR). The two brackets are orthogonal. The δ on the left-hand side is the global quantity of change pervading the whole system (system time: 1 cycle = 1 tick), while the time axis on the right-hand side is the local domain axis inside the classical bracket. The present paper additionally uses the fact that structure (the registration of brackets and axes) and cost (the payment of boundary crossings) are registered in different lists - see the number registry of Section 2.4.

Axiom 2 (The conditional state transition (CAS) is the unique operator). *Every change of the universe is the repetition of a single conditional state transition. CAS proceeds in 3 mutually orthogonal steps - observe (Read), compare (Compare), record (Swap). Fewer than 3 steps violates causality; more than 3 violates least cost.*

Axiom 3 (Cost). *Crossing an orthogonal boundary + in the prescribed order incurs cost +1; without order the cost is 0. Cost is the sole physical quantity of δ - energy, mass, force, and entropy are all different names for cost (proposition of Axiom 4 of the complete system [6]).*

Axiom 4 (δ is the global flag). *δ is a single global firing flag outside the finite state machine; it corresponds to Axiom 15 of the complete system [6].*

Proposition 1 (Complete-description degrees of freedom 7). *The domain 4 axes and the CAS 3 steps are mutually orthogonal, so the internal degrees of freedom of the system number $7 = 4 + 3$.*

We recall two definitions (Paper I Section 2). *Definition 1: total cost $13 = \text{read cost } 8 + \text{write cost } 5$. Definition 2 (ball value): $3\text{-axis grip} + \text{time} = 1\text{ time} + 3\text{ space} = 4$. The residual cost is $13 - 4 = 9$.*

Following the way Paper II first formalized its correspondence rule as Definition 3 before stating its theorems, we state as working definitions the two correspondence rules the present derivations use (numbered Definitions 4 and 5, continuing Definitions 1, 2 of Paper I and Definition 3 of Paper II).

The two definitions are counting rules induced by the cost table (Axiom 4) and the number registry of the complete system [6]; they are not new axioms.

Definition 4 (Context-bracket correspondence). *If the context of a physical quantity is static (storage of state, preservation of records), its structural constant is counted from the registered numbers of the structural bracket (DATA); if dynamic (emission, damping, payment), from the registered numbers of the cost bracket (OPERATOR). Ground: the orthogonality of the two brackets in Axiom 1 - the separation between where records are preserved (DATA) and where operations are paid (OPERATOR) forces the separation of contexts.*

Definition 5 (Partial-sum accounting). *Cost components are counted as the partial sum the context requires. (a) Spatial-contraction context: only the state-changing cost-bearing steps are counted. Of the three CAS steps, Read is an observation that changes no state, so the steps counted toward contraction are Compare and Swap, i.e. 2. (b) Domain-emission context: only the components crossing a domain boundary are counted. Of the write cost 5, the DATA-internal commit 1 crosses no domain boundary, so the partial sum counted toward emission is the ball value $4 = 1\text{ time} + 3\text{ space}$ (Definition 2). Ground: the cost-incurrence condition of Axiom 3 (boundary crossing) and the component decomposition of the cost table of the complete system [6]. The total accounting of the complete system (R, C, S each +1; write 5) is preserved unchanged; this definition fixes only which partial sum corresponds to which physical context.*

2.3. Exhaustive List of Items Used in This Paper

Table 1. Exhaustive list of the axioms, proposition, definitions, and earlier-paper theorems used in this paper. E1 through E3 are complete-system [6] items used for the first time in this paper.

ID	Item	Content	Used in
A1	Axiom 1 (Banya equation)	4 orthogonal axes, two brackets	throughout
A2	Axiom 2 (CAS 3 steps)	Read, Compare, Swap	Thm. 4, Sec. 7
A3	Axiom 3 (cost)	boundary crossing +1, cost sole quantity	Thm. 3, 4, Sec. 8
A4	Axiom 4 (δ flag)	single global firing bit	Sec. 7
P1	Proposition 1	$7 = 4 + 3$	Thm. 3
D1	Definition 1	$13 = 8 + 5$	Sec. 7
D2	Definition 2 (ball value)	$4 = 1 \text{ time} + 3 \text{ space}$	Thm. 1, 2
E1	number registry (complete system Axiom 9)	two DOF lists, structural/cost	Thm. 1, 3, App. B
D4	Definition 4 (context-bracket)	static = structural, dynamic = cost	Thm. 1, 2
D5	Definition 5 (partial-sum)	contraction = 2 steps, emission = ball value 4	Thm. 1, 2, 4
E2	recovery rule (complete system Axiom 6)	RLU cost recovery	Thm. 3
E3	juim recording (complete system Axiom 7)	written (gripped) states recorded in d-ring slots	Thm. 1, 4
T1	Paper I [5] Theorem 1	forces (5, 2)	Thm. 3 (input)
T2	Paper II [7] Theorem 1	bit reading (0, 1, 0, 1)	Thm. 2 (input)

2.4. The Number Registry (E1)

Axiom 9 of the complete system [6] registers, in two lists, the numbers the system may use in derivations. Structural degrees of freedom (classical bracket, DATA): 1 (bit basis), 2 (two brackets), 3 (CAS 3 steps), 4 (four domains), 7 (internal DOF), 9 (structural complete descriptive DOF = $7 + 2$), 16 (2^4 domain combinations), 30 (access paths), 128 (2^7 valid states), 137 (data type maximum). Cost degrees of freedom (quantum bracket, OPERATOR): 1 (cost basis), 2 (absent irreversible axes), 3 (CAS internal cost), 4 (ball value), 5 (five irreversible axes), 9 (residual cost = $13 - 4$), 13 (total cost). Factorable numbers (6, 8) and norm derivatives ($\sqrt{1}$, $\sqrt{2}$, $\sqrt{3}$) are explicitly excluded, and the filter rule drops 10 as an intermediate step.

The decisive fact for this paper is this: 4 is registered in both lists separately. The 4 of the structural list is the domain count; the 4 of the cost list is the ball value. Since the two brackets are orthogonal by Axiom 1, the two 4s are different quantities sharing a numerical value. The derivation notation of the registry (e.g. $9 = 7 + 2$) describes why each number is registered and is a different tier from the cross-bracket combination rule [6]. Appendix B checks every number appearing in this paper against the registry.

2.5. Terminology

Juim (grip): one occupancy of state, confirmed by writing (gripping) and recorded in a d-ring slot (complete system Axiom 7). *d-ring*: the recording slot ring of DATA. *RLU*: the cost-recovery replacement rule (complete system Axiom 6). *Ball value*: the cost $4 = 1 \text{ time} + 3 \text{ space}$ paid by one write (Definition 2). *Bit reading*: the 4-bit value reading the on/off of the four domain axes (Paper II Section 2).

3. Theorem 1: Bracket Separation of the Entropy Coefficient 4

3.1. Statement

Theorem 1 (Bracket separation of the two 4s). Under the counting rules of Definitions 4 and 5, the coefficient 4 of the Bekenstein-Hawking entropy $S = A / (4\ell_P^2)$ is the structural degree of freedom 4 (domain count, Axiom 1), while the 4 of the QNM asymptotic damping spacing $M\Delta\omega_I = 1/4$ is the cost degree of freedom 4 (ball value, Definition 2). The two 4s are different quantities registered in different brackets of the number registry (E1), coinciding numerically in three spatial dimensions. [Class I - under Definitions 4, 5]

3.2. Proof

(A) Registration of the two 4s. The number registry (Section 2.4) registers 4 in the structural list (four domains: observer, superposition, time, space) and 4 in the cost list (ball value: 3-axis grip + time) separately. By Axiom 1, structure (registration of brackets and axes) and cost (payment of boundary crossings) belong to orthogonal brackets, so equal numerical values are nonetheless different quantities. This is a direct application of filter rule 3 of Paper I [5] Section 8.2 (same numeral, different category, different quantity).

(B) The static context is the structural 4. Entropy is the information storage per area. Storage is the recording of state, not the payment of cost, so by Definition 4 its structural constant is counted from the structural bracket. By juim recording (E3, Axiom 7) juims are recorded in d-ring slots, and the minimal partition occupying one slot is the 4-axis orthogonal partition (Axiom 1). The holographic principle - bulk degrees of freedom reduce to the boundary [23,24] - appears in this system as the fact that CAS interacts only at the ring seams of the domain boundary; since the minimal unit occupying one area bit is the 4-axis partition, the denominator 4 is forced. Hence the 4 of the storage context (area entropy) is the domain count:

$$S = \frac{A}{4\ell_P^2}, \quad 4 = \text{domain count (structural bracket)}. \quad (5)$$

(C) The dynamic context is the cost 4. QNM damping is an emission process, so by Definition 4 its structural constant is counted from the cost bracket. In the asymptotic regime one damping step corresponds to one domain emission, and by Definition 5(b) the counted partial sum of one emission is the ball value $4 = 1 \text{ time} + 3 \text{ space}$. The payment unit being 4, the damping spacing is its inverse:

$$M\Delta\omega_I = \frac{1}{4}, \quad 4 = \text{ball value (cost bracket)}. \quad (6)$$

This value is identical (0% error) to the exact $1/4$ of Nollert's numerics [16] and Motl's analytic proof [15].

(D) Discrimination: higher-dimensional invariance. If the two assignments are correct, a divergence is predicted. Naively extrapolating the components of the ball value gives $1 + d$ in spatial dimension d (time 1 + space d). But it is a known literature fact that the entropy coefficient of higher-dimensional holography, $S = A / (4\ell_P^{d-1})$, remains 4 independently of dimension - the domain count is the number of axes of Axiom 1, not the number of spatial dimensions, so the present assignment concords with this invariance. Hence the entropy 4 is the domain count, not the ball value, and the two 4s coincide numerically only at $d = 3$.

3.3. Remark - What This Theorem Establishes

What this theorem establishes is the *separation, not the isomorphism*, of the two 4s. The registry lists the two 4s in different brackets; the theorem fixes which context corresponds to which 4. The numerical coincidence is an accident of $d = 3$, and the separation surfaces in higher dimensions. Treating equal

numerals as categorically distinct, with the divergence testable - this is what distinguishes the present derivation from numerology.

Notice of change against the source card. The source card H-979 recorded this object as the hypothesis “domain 4 = ball value 4 isomorphism” and left the axiomatic proof of the isomorphism open. The present theorem inverts that frame: since the registry lists the two 4s in different brackets, there is no isomorphism to prove, and what remains to be fixed is the context assignment. This inversion (from isomorphism hypothesis to bracket separation) is made in the present paper, and the card is revised accordingly. The observation that read the two 4s as two faces of one structure (equal numerals, $d = 3$) is explained at its origin by part (D).

Comparison with higher-dimensional asymptotic QNMs. The asymptotic damping spacing of higher-dimensional Schwarzschild-Tangherlini black holes is already known analytically: spacing $= 2\pi T_H$ [26,27]. At $d = 3$, $2\pi T_H = 1/(4M)$, in exact agreement with part (C). In higher dimensions this temperature formula and the naive component extrapolation $1/(1 + d)$ give different values, so the higher-dimensional extension of the ball-value reading is an open item requiring the explicit construction of a d -dimensional cost accounting (reconstruction of the read and write components; Table 4). Part (C) is therefore settled at $d = 3$, and the higher-dimensional discrimination is decided on top of that construction (Prediction 1 of Section 10). The entropy side of the discrimination (dimension invariance of the coefficient 4) is already supported by the known literature and is classified as post-hoc support, not as a prediction. Figure 1 summarizes the assignment.

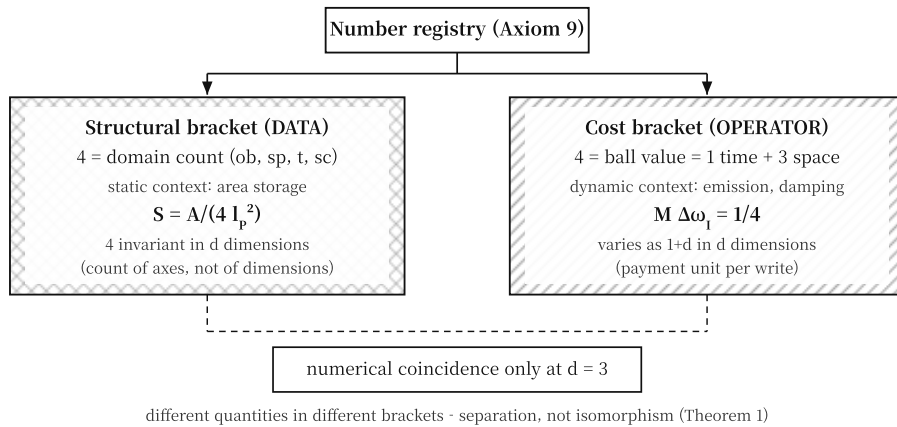


Figure 1. Bracket separation of the two 4s (Theorem 1). From the number registry the structural bracket (domain 4, static context, area storage, dimension-invariant) and the cost bracket (ball value 4, dynamic context, emission and damping, varying as $1 + d$) branch apart, coinciding numerically only at $d = 3$. Cross-hatch marks the structural bracket, hatch the cost bracket.

4. Theorem 2: The Wheeler-DeWitt Lattice Injection Is the Imprint of the Empty Time Axis

4.1. Statement

Theorem 2 (Imprint of absent time). *In the bit reading (observer, superposition, time, space) = $(0, 1, 0, 1)$ of Paper II [7] Theorem 1, time = 0, so the ball value 4 = 1 time + 3 space cannot be procured inside the Wheeler-DeWitt slice. Hence obtaining the entropy coefficient 1/4 from this equation forces a discrete scale to be injected from outside the equation. [Class I - under Definitions 4, 5]*

4.2. Proof

(i) By Paper II Theorem 1, the four constituents of the Wheeler-DeWitt equation $\hat{H}\Psi[h_{ij}] = 0$ [4] force the domain bit reading uniquely to $(0, 1, 0, 1)$, the remaining 15 combinations being exhaustively excluded (Paper II Appendix A). In this reading, time = 0.

(ii) By Definition 2 the ball value 4 consists of 1 time + 3 space; its completion necessarily includes the time component 1.

(iii) On a slice with time = 0 the time component cannot be paid, so the ball value 4 is not completed.

(iv) By Theorem 1 (B) the entropy coefficient 4 is the domain count, but the *process of producing* the entropy (the counting of microstates) requires the discrete unit of writing - microstates are counted in units of juims (Section 2.3 terminology), juims are recorded by writing (E3, Axiom 7), and by Definition 5(b) the emission counting unit of writing is the ball value 4. Since that unit cannot be procured inside the slice, the discrete scale can only be injected from outside the equation.

4.3. Post-Hoc Concordance - Vaz's Discrete Spectrum

Vaz's canonical quantization of the Schwarzschild black hole [8] had to introduce a Bekenstein-type discrete mass spectrum $M \sim \sqrt{N} M_P$ in order to derive the statistical entropy from the time-eliminated constraint equation - the discrete spacing comes from an external assumption, not from the equation itself. The present theorem explains why that methodological choice was unavoidable: the time = 0 slice is a place where self-procurement of the discrete unit is structurally blocked, and the spectrum injection is the imprint of that empty place. This is no defect of that literature - on the contrary, the content of the theorem is precisely that the methodological constraint of that work meshes with the structure of the present system.

Significance and qualification. The bit reading of Paper II here shows, for the first time, explanatory power beyond postdiction. If the reading is right, the ball value must be unprocurable on the time = 0 slice - and indeed the independent literature injected an external discrete spectrum. This correspondence was explained after the fact, not predicted, so it is a single post-hoc concordance and we claim no more. We also state the distinction from a mundane alternative explanation: the general fact that counting statistical microstates requires a discrete structure, independently of time elimination, could by itself account for the spectrum injection. The place where the two explanations part is a canonical formalism in which time survives - if the mechanism of the present theorem is right, the same external injection must not be forced in a formalism with time switched on. We formalize this discrimination as the contrast clause of Prediction 4 in Section 10; until that contrast is carried out, the status of this subsection remains a concordance case. Figure 2 summarizes the argument.

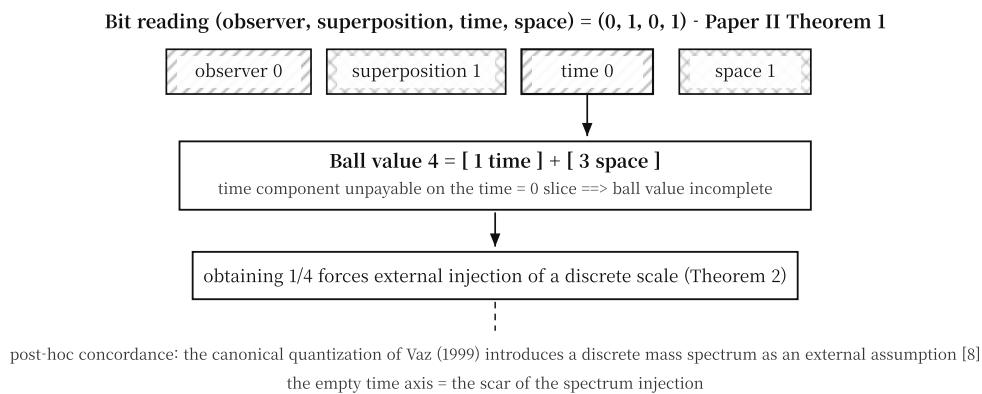


Figure 2. At the time = 0 position of the bit reading (0, 1, 0, 1), the time component of the ball value 4 is unprocurable (Theorem 2), so obtaining 1/4 forces the external injection of a discrete scale. Vaz's discrete-spectrum introduction [8] is its imprint. Cross-hatch marks axes that are on (1), hatch axes that are off (0).

5. Theorem 3: The Temperature-Lifetime Identity and the Evaporation Coefficient

5.1. Statement

Theorem 3 (Temperature-lifetime identity). *The Hawking temperature and evaporation lifetime of a Schwarzschild black hole satisfy the identity*

$$T_H^3 \cdot \tau_{BH} = \frac{10}{\pi^2} T_P^3 t_P. \quad (7)$$

The mass dependence cancels exactly as $M^{-3} \times M^3$ (an identity by definition). The evaporation coefficient decomposes as $5120 = 10 \times 2^9$, where $10 = \dim SO(5)$ is the group dimension of the signature $(5, 2)$ forced by Paper I [5] Theorem 1 (class II), and the 9 of 2^9 is the structural complete descriptive DOF $9 = 7 + 2$ (class I). [Class I + II]

5.2. Computation

Starting from the standard expressions [2,3],

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}, \quad \tau_{BH} = \frac{5120 \pi G^2 M^3}{\hbar c^4}, \quad (8)$$

cubing and multiplying cancels $M^{-3} \times M^3$ and

$$T_H^3 \tau_{BH} = \frac{5120 \pi}{(8\pi)^3} \frac{\hbar^2 c^5}{G k_B^3} = \frac{5120}{512 \pi^2} T_P^3 t_P = \frac{10}{\pi^2} T_P^3 t_P = 1.013212 T_P^3 t_P \quad (9)$$

($T_P^3 t_P = \hbar^2 c^5 / (G k_B^3)$) follows automatically from the definitions of the Planck temperature and time - step-by-step expansion in Appendix A). The identity holds for every Schwarzschild black hole; it is not an approximation. We separate the statuses: the identity (7) itself is an algebraic rearrangement of two standard formulas and holds independently of the axioms. The axiomatic content of the theorem is the structural decomposition of the coefficient ($5120 = 10 \times 2^9$) and the tier assignment of its two factors. Rotating and charged black holes require corrections, which correspond to additional juim cost on the d-ring (Section 9).

Reading inside the system. A black hole is the state in which juims fill the d-ring, and Hawking radiation is the recovery of juims under the recovery rule (E2). The product of the cubed temperature and the lifetime is the invariant in which this recovery cost closes independently of mass: the denser the juims (the larger the mass), the lower the temperature and the longer the lifetime, but the cube-product is invariant.

5.3. The Status of 10

10 is not a degree of freedom of the number registry. The filter rule of the complete system [6] explicitly drops 10 as an intermediate step. In this paper 10 is the dimension of $SO(5)$, the maximal compact subgroup of the preservation group $SO(5, 2)$ of the signature $(5, 2)$ forced by Paper I [5] Theorem 1 - a number that mathematics fixes automatically once the axioms force the signature (class II). Paper I treated $8\pi^4$ at the same status - "once $(5, 2)$ is forced, mathematically determined from the group volume of $SO(5) \times SO(2)$ " - and the same Paper I Table 5 put the residual cost 9 in analytic correspondence with $\dim SO(5) - \dim SO(2) = 10 - 1$. The DOF filter is a rule for the data type; group-theoretic constants belong to a different tier. This tier distinction is spelled out exhaustively in the checklist of Appendix B.

Series connection. The 10 of the black-hole evaporation coefficient and the $\dim SO(5)$ of the fine-structure derivation (Paper I Section 4) are the same number. The two physical quantities descend from the same signature $(5, 2)$ by different routes. This is what Theorem 3 means for the series: the

group dimension that produced the electromagnetic coupling constant also appears in the black-hole evaporation rate.

Reading of 2^9 . 9 is registered in the structural list of the registry as the structural complete descriptive DOF $9 = 7 + 2$ (internal DOF 7 and brackets 2; class I), and $2^9 = 512$ is its binary state count. The evaporation coefficient splits into group structure (dimension 10) and information state space (2^9), the two factors forced on different grounds. The source card H-54 read the 9 as “CAS 3 + domain 4 + firing 2,” but a firing 2 bits conflicts with Axiom 4 (a single δ flag); the present paper adopts the registry reading $9 = 7 + 2$, and the card is revised accordingly.

Uniqueness of the decomposition. In a decomposition $5120 = 2^a \times b$, the exponent a is the bit count of a binary state space and must be a registered degree of freedom, while the factor b must be either a registered number or of group-theoretic tier (class II). Enumerating the candidates: 5×2^{10} is excluded because the exponent 10 is an explicitly dropped number (intermediate step); 20×2^8 is excluded because the exponent 8 is explicitly dropped (2^3 factorable); 40×2^7 has a registered exponent 7 but is excluded because the factor 40 is neither registered nor group-theoretic. The surviving decomposition is 10×2^9 (registered exponent 9, group-tier factor 10) - unique.

6. Theorem 4: Contraction Stacking of the Event-Horizon Radius

6.1. Statement

Theorem 4 (Contraction stacking). *Reading the mass M as the juim count $N = M/m_P$ in Planck-mass units, and counting the contraction unit of one juim as the $2\ell_P$ corresponding to the two cost-bearing steps of Definition 5(a), the stacking radius of N juims is*

$$r_s = N \times 2\ell_P = \frac{2GM}{c^2}, \quad (10)$$

reproducing the standard Schwarzschild radius (an identity by definition). [Class I - under Definition 5]

6.2. Proof Sketch

That the contraction unit of one juim is $2\ell_P$ follows from Definition 5(a): in the spatial-contraction context only the state-changing cost-bearing steps are counted, and of the three CAS steps those are Compare and Swap (Read is an observation changing no state). The factor 2 is not an arbitrary constant but the consequence of this counting rule. Substituting $N = M/m_P$, $\ell_P = \sqrt{\hbar G/c^3}$, $m_P = \sqrt{\hbar c/G}$ yields $r_s = 2GM/c^2$ as an identity (Appendix A.4). The formal content of the theorem is this contraction-stacking identity; the exactness of the reproduction is a property of the identity, not independent evidence.

The cumulative-cost picture (interpretation, class III). Card D-49 reads this boundary as the saturation surface at which the pairwise cumulative CAS cost of N juims, $N(N-1)/2 \approx N^2/2$, reaches the escape energy and the d-ring closes. This picture is the interpretive layer wrapping the identity of the present theorem; until an in-system quantitative definition of the escape energy is supplied, it is confined to class III.

The difference of picture. In this account, information fails to escape not because the Read cost diverges but because *the Read path itself is blocked*. Inside the boundary where the N^2 accumulation exceeds the escape energy, the d-ring closes (saturation of juim recording, E3) and observation operations from outside become impossible. Saturation, not divergence. The result agrees with the standard account (causal isolation of signals inside the horizon) while the mechanism pictured differs, and this difference concords with the storage/emission distinction of Theorem 1: the horizon is the boundary where storage saturates, and Hawking radiation is the recovery rule beginning its recovery at that boundary (Theorem 3).

7. Auxiliary Results: Loop Quantum Gravity and Quasinormal Modes (Class III)

The gamma, the area gap, and the QNM real part of this section are results already published in Section 7.1 of Paper II [7]. This section does not re-derive them but uses them as input; what is new is twofold: the ball-value reading of the QNM imaginary spacing (re-confirmation of Theorem 1 (C)), and the ratio $3\pi/8$ of the two area quanta.

7.1. Numerical Summary

The Barbero-Immirzi parameter [19,20] is the sole free parameter of loop quantum gravity, fixed by hand; the area gap is the standard formula $\Delta A = 8\pi\gamma\sqrt{j(j+1)}\ell_p^2$ at the minimal representation $j = 1/2$ [18].

Table 2. Auxiliary numerical results, all of class III (axiom-described). The gamma and the area gap are results of Paper II Section 7.1 and are not re-derived here.

Quantity	Closed form	Derived	Literature	Deviation
Barbero-Immirzi γ	$\sqrt{3} \ln 3/8$	0.2378565	0.23753296 (transcendental root [13,14])	0.136% (definite)
LQG area gap ΔA	$(3\pi/2) \ln 3 \ell_p^2$	$5.177088 \ell_p^2$	$4\pi\sqrt{3} \gamma_M \ell_p^2$ $5.170046 \ell_p^2$	= 0.136% (γ -dependent)
QNM real part $M\omega_R$	$\ln 3/(8\pi)$	0.04371239	0.0437123 [15,16]	0.0002%
QNM imaginary spacing $M\Delta\omega_I$	inverse ball value $1/4$	0.25	$1/4$ (Motl exact [15])	0% (identity)
LQG gap / Hod quantum	$\frac{(3\pi/2) \ln 3}{4 \ln 3} = \frac{3\pi}{8}$	1.178097	identity	0%

7.2. The Ratio $3\pi/8$ of the Two Area Quanta

The original attempt to connect QNMs with the LQG area quantum is Dreyer's [28]; Hod obtained, by a Bohr-correspondence argument on the QNM real part, the area quantum $\Delta A_{\text{Hod}} = 4 \ln 3 \ell_p^2$ [17]. This quantum reads as the product of the entropy denominator 4 (domain count, Theorem 1) and the CAS three-state information $\ln 3$ - the domain count and the CAS information cohabit in one formula. On the imaginary-spacing side there is the competing reading of Maggiore [29], who read the spacing as a physical frequency and proposed the area quantum $8\pi\ell_p^2$ - the same numerical object as the present reading (inverse ball value) with a different interpretation; the discrimination is tied to the d -dimensional accounting (Prediction 1 of Section 10). Taking the ratio with the LQG area gap (Table 2), $\ln 3$ cancels and

$$\frac{\Delta A_{\text{LQG}}}{\Delta A_{\text{Hod}}} = \frac{(3\pi/2) \ln 3}{4 \ln 3} = \frac{3\pi}{8} \quad (11)$$

remains. The famous mismatch of the two camps' area quanta (LQG vs. the QNM heuristic) reduces, in this system, to the single geometric factor of the hemispheric three-axis arc 3π divided by the ring 8 bits. The mismatch is not resolved; its magnitude is specified as a ratio of structural constants.

7.3. The Tier of 8 and $\sqrt{3}$ - Why This Section Is Class III

The components $\sqrt{3}$ and 8 of the gamma are not registry degrees of freedom: the registry explicitly excludes norm derivatives (including $\sqrt{3}$) and factorable numbers ($8 = 2^3$). Both quantities are defined inside the system - $\sqrt{3}$ is the workbench three-axis norm (proposition of Axiom 2) and 8 is the d-ring 8 bits of the data type $137 = 8 + 128 + 1$ (Axiom 4) - but they are not of registry tier, so the results of this section are not registered-number combinations and are confined to class III (axiom-described). This demotion is not concealment of a defect but the filter at work: a case in which the registry filter screens the system's own results, presented as evidence that the rules are not ornamental. The candidate-search

record is also disclosed, as in Paper II Section 7.1: before fixing the adopted form, card D-159 tried $3/(4\pi)$ (deviation 0.51%) and $3 \ln 2 / (2\pi\sqrt{2})$ (1.48%) and adopted the closed form nearest the literature value. This target-matched selection is part of the reason this section is confined to class III. The origin of the 0.136% deviation (the literature value is the root of a transcendental equation with an infinite spin sum; the derived value is a closed form from a finite ring truncation) is an open item of Section 9.

8. Position in the Lineage: Jacobson, Padmanabhan, Verlinde

With the thermodynamics of information and computation [21,22] and the holographic principle [23,24] as headwaters, the deep relation between gravity and thermodynamics is represented by a lineage of three names. Jacobson derived the Einstein equation as an equation of state by applying the Clausius relation $\delta Q = T dS$ to local causal horizons [9]. Padmanabhan showed that the field equations evaluated on a horizon take the form of the first law of thermodynamics, and systematized this [10]. Verlinde cast gravity itself as an entropic force [11]. The lineage is still advancing: the most recent constraint-free formulation [25] takes the entropy variation instead of the volume variation and derives the first law without extra constraints even for multi-horizon black holes. What deserves notice is that even with the derivation format thus refined, the 4 of $S = A/4$ and $T = \kappa/2\pi$ remain inputs assumed by standard identification - the refinement of format and the origin of the constants are problems of different tiers.

Table 3. Tier contrast between the lineage and the present system: what each line takes as input and what it produces.

Line	Takes as input	Produces
Jacobson [9]	$S \propto A$, Unruh temperature	Einstein equation (as equation of state)
Padmanabhan [10]	field equations, $T = \kappa/2\pi$, $S = A/4$	formal equivalence with the first law
Verlinde [11]	holographic principle, equipartition	Newton's laws, inertia (as entropic force)
Constraint-free formulation [25]	field equations, $T = \kappa/2\pi$, $S = A/4$	constraint-free first-law format for multi-horizon black holes
Present series	4 axioms + 1 proposition + Definitions 4, 5	the bracket assignment of 4 and 1/4, the structural decomposition of 5120, the contraction identity of r_s

The present system shares the direction of this lineage but differs in tier. The lineage takes $S \propto A$ and the horizon temperature as inputs and shows the equivalence of equations and thermodynamics. The present system assigns the structural origin of the coefficients of those inputs - the bracket assignment of the 4 of $S = A/4$ (Theorem 1), the structural decomposition of the evaporation coefficient 5120 (Theorem 3), the contraction identity of the horizon radius (Theorem 4). It works one floor below where that lineage stands.

The difference from Verlinde's entropic gravity is more fundamental. Verlinde cast gravity as an entropic force, and this casting received a criticism grounded in ultracold-neutron experiments [12]. In the present axiomatic system, force is not an independent fundamental quantity - energy, mass, force, and entropy are all different names for cost (Axiom 3). Since force does not stand as a fundamental quantity, the very tier at which gravity could be cast as an entropic force does not exist in this system, and the criticism of entropic force therefore does not apply. Here the horizon is not an equilibrium surface of forces but a saturation boundary of cumulative cost (interpretation of Theorem 4), and Hawking radiation is not the effect of an entropic force but the recovery process of the recovery rule (Theorem 3).

9. Discussion: Scope and Limits

Self-notice of retrospective derivation. The derivations of Sections 3 through 7 are post-hoc reconstructions of known facts ($S = A/4$, $1/4$, 5120, r_s , γ) and carry no predictive risk by themselves.

The predictive risk is carried by the forward items of the table in Section 10. This division is the same as in Paper II [7].

Table 4. Exhaustive list of open items. Class III: derivation path and interpretation in the text, independent formal proof deferred. IV: outside the scope of this paper.

Class	Item	Current state	Direction
III	γ 0.136% definite deviation	literature value is a transcendental root of an infinite spin sum; derived value is a finite ring truncation	re-derivation of the infinite sum under ring-cyclic truncation
III	the 1/2 of the QNM offset $n + 1/2$	candidate: Compare binary-branch half-tick	derivation open
III	d -dimensional cost accounting of the ball value	naive component extrapolation $(1 + d)$ disagrees with the known temperature formula $2\pi T_H$ [26,27] (Section 3.3)	reconstruct the read and write components in d dimensions, then decide by Prediction 1
IV	multiple horizons ($r_{\pm}$), rotation and charge corrections	the constraint-free first-law format was settled by [25]; the question of this paper is the origin of the constants appearing in that format, and in the multi-horizon extension the corrections correspond to additional juim cost on the d-ring	subsequent paper
IV excluded	higher-derivative gravity the 2π of the Bekenstein bound	- the in-system role of $\hbar c$ unresolved	out of scope not treated in this paper

Physical status of the evaporation coefficient. The 5120 of Eq. (2) is the standard coefficient of the photon black-body emission approximation. Calculations that include greybody factors and particle species precisely [3] change the effective coefficient. The claim of this paper is not the numerical value of an observed lifetime but the structural decomposition 10×2^9 of the black-body coefficient 5120 and the identity (7). Prediction 3 (micro black holes) must be read under this qualification.

The circularity concern. The answer to “the axioms have the answers built in” is the same as Paper I [5] Section 8.2 and is not repeated here. The gist: the number registry has never been adjusted retroactively to fit a result (see the prior record in Section 1 - the version history of the registry itself is traceable through the versioned records of the complete system), and results that use numbers absent from the registry $(10, 8, \sqrt{3})$ state that fact and lower their tier (Appendix B). The two interpretation changes between cards and this paper (H-979 from isomorphism to separation; the 9-reading of H-54) are disclosed in full at the relevant sections (Sections 3.3, 5.3).

10. Conclusion

A third physical domain has come out of the same 4 axioms and 1 proposition. Paper I [5] obtained the electromagnetic coupling constant $\alpha = 1/137.036\,082$ from the signature $(5, 2)$; Paper II [7] obtained the DeWitt metric's $(5, 1)$ from the same $(5, 2)$; the present paper obtained the three structural constants of black hole thermodynamics from the same structure: the 4 of $S = A/4$ is the domain count (Theorem 1), the 1/4 of QNM damping is the inverse ball value (Theorem 1), and the 10 of the evaporation coefficient $5120 = 10 \times 2^9$ is Paper I's own $\dim SO(5)$ (Theorem 3). That the two 4s are different quantities in different brackets, and that their separation is testable in higher dimensions, is the central result. The bit reading of Paper II explains the methodological constraint of an independent literature (discrete-spectrum injection) structurally, showing for the first time explanatory power beyond postdiction (Theorem 2). The event-horizon radius is reproduced as an identity by contraction stacking, wrapped by the picture of cumulative-cost saturation (class III) (Theorem 4).

Table 5. Status tally of the results of this paper.

Class	Items	N
I. Axiom-derived (under Def. 4, 5)	Theorem 1 (bracket separation), Theorem 2 (spectrum injection), the 2^9 assignment of Theorem 3, Theorem 4 (contraction identity)	4
standard algebra (axiom-free)	the identity (7) of Theorem 3	1
II. Axiom-determined	the $10 = \dim SO(5)$ of Theorem 3	1
III. Axiom-described	γ , area gap, QNM real part, $3\pi/8$, offset $1/2$, cumulative-cost picture (interpretation of Theorem 4)	6
IV. Out of scope	multiple horizons, higher-derivative gravity	2

Table 6. Unique predictions and falsification conditions. Items without a falsification condition are not listed; the item already supported by the known literature (dimension invariance of the entropy coefficient) is classified as post-hoc support, not prediction, and is excluded here (Section 3.3).

No.	Prediction	Ground	Falsification condition
1	once a d -dimensional cost accounting is constructed, its higher-dimensional asymptotic QNM spacing must agree with the known temperature formula $2\pi T_H$ [26,27] (conditional)	Theorem 1 (C) and the literature contrast of Section 3.3	if the constructed accounting gives a spacing different from the temperature formula, the ball-value assignment is falsified
2	loss of $\ln 3$ in non-isotropic (Kerr / Reissner-Nordstrom) collapse	Section 7: $\ln 3$ is the CAS three-state information of isotropic collapse. The RN asymptotics were already computed by [26], where pure $\ln 3$ does not survive, concordant with this prediction (an exhaustive Kerr contrast is subsequent)	if $\ln 3$ survives in non-isotropic collapse, falsified
3	the black-body-approximation coefficient of micro-black-hole evaporation is $5120 = 10 \times 2^9$ (a prediction shared with standard physics; low discriminating power)	Theorem 3	if the observed coefficient at the same level of approximation departs from 5120, falsified
4	every derivation of $S = A/4$ in a time-eliminated formalism requires an external discrete scale. External means the spectral spacing is not derived from the solutions of the constraint equation but introduced as a separate assumption; as a contrast clause, the same requirement must not be forced in a canonical formalism with time switched on	Theorem 2	if $S = A/4$ emerges from the Wheeler-DeWitt equation without an external scale, falsified; if the same requirement is universal even with time on, the specificity of this mechanism is falsified

Of the two axes of the higher-dimensional discrimination, the entropy side (dimension invariance of the coefficient 4) is already supported by the known literature and is post-hoc support; the QNM side remains as Prediction 1: the moment a d -dimensional cost accounting is constructed and contrasted with the known temperature formula, the life or death of the ball-value assignment is decided. Prediction 4 is methodological, adjudicated every time a new computation appears in a time-eliminated formalism.

The system-level falsification condition is shared across the series: the unique predictions of Paper I [5] (e.g. falsified if the nearest integer to $1/\alpha(M_Z)$ departs from 128), while the predictions of this paper aim at the assignment level. The full list of unique predictions of the complete system can be read online (<https://ubmscoin.github.io/banya/en/predictions.html>).

The chain of the series now has three branches: from one (5,2) came the electromagnetic coupling constant (Paper I), the metric of gravitational quantization (Paper II), and the structural constants of black hole thermodynamics (this paper). Figure 3 summarizes the chain. The next tasks are the open items of Table 4 in Section 9.

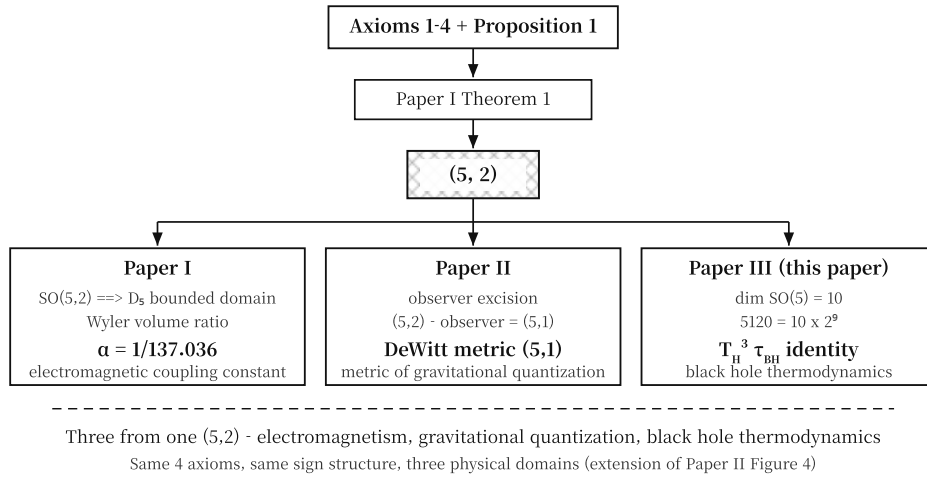


Figure 3. The three-branch chain of the series (extension of Paper II Figure 4). Axioms 1-4 and Proposition 1 $\Rightarrow$ Paper I Theorem 1 $\Rightarrow$ (5, 2) $\Rightarrow$ three branches: Paper I $SO(5,2) \Rightarrow D_5$ Wyler volume ratio $\Rightarrow \alpha$ / Paper II observer excision $\Rightarrow$ (5, 1) DeWitt metric / this paper $\dim SO(5) = 10 \Rightarrow 5120$ temperature-lifetime identity. Cross-hatch marks the shared knot (5,2).

Table 7 compares the scope of derivations of the present paper (4 axioms + 1 proposition) with the complete axiomatic system [6] (15 axioms). The present paper treats only the minimal axiom set needed for the three structural constants of black hole thermodynamics.

Table 7. Scope of derivations - the present paper and the complete axiomatic system.

Item	This paper	Complete system [6]
Number of axioms	4 + 1 prop.	15
Derivation items (Table 5 classification), subtotal	14	1,096
Unique predictions (experimentally testable, Table 6)	4	130

Appendix A Numerical Evaluation

A.1. Step-by-step expansion of the identity (7). From Eq. (8),

$$T_H^3 \tau_{BH} = \left(\frac{\hbar c^3}{8\pi G k_B} \right)^3 M^{-3} \times \frac{5120 \pi G^2 M^3}{\hbar c^4} = \frac{5120 \pi}{512 \pi^3} \frac{\hbar^3 c^9}{G^3 k_B^3} \frac{G^2}{\hbar c^4} = \frac{10}{\pi^2} \frac{\hbar^2 c^5}{G k_B^3}. \quad (A1)$$

From the Planck temperature $T_P = \sqrt{\hbar c^5 / (G k_B^2)}$ and Planck time $t_P = \sqrt{\hbar G / c^5}$,

$$T_P^3 t_P = \left(\frac{\hbar c^5}{G k_B^2} \right)^{3/2} \left(\frac{\hbar G}{c^5} \right)^{1/2} = \frac{\hbar^{3/2} c^{15/2}}{G^{3/2} k_B^3} \frac{\hbar^{1/2} G^{1/2}}{c^{5/2}} = \frac{\hbar^2 c^5}{G k_B^3}, \quad (A2)$$

so (A1) and (A2) agree and Eq. (7) holds. $10/\pi^2 = 1.013212$.

A.2. Damping spacing and real part. $M\Delta\omega_I = 1/4 = 0.25$ (identical to Motl's exact value). $M\omega_R = \ln 3/(8\pi) = 1.0986123/25.132741 = 0.0437124$ (0.0002% against Nollert's numerical 0.0437123).

A.3. Gamma and area gap. $\gamma = \sqrt{3} \ln 3/8 = 1.7320508 \times 1.0986123/8 = 0.2378565$; against the literature transcendental root 0.23753296 [13,14] the deviation is 0.136% - since the literature value is an exact root without error bars, this is a definite deviation, not a statistical agreement. $\Delta A = 4\pi\sqrt{3}\gamma\ell_p^2 = (3\pi/2) \ln 3 \ell_p^2 = 5.177088 \ell_p^2$; the $\sqrt{3}$ inside γ and the $\sqrt{3}$ of the minimal-representation Casimir $\sqrt{3}/2$ multiply to the integer 3 and the radical cancels. Ratio $(3\pi/2) \ln 3/(4 \ln 3) = 3\pi/8 = 1.178097$.

A.4. Schwarzschild radius. $r_s = 2N\ell_p = 2(M/m_P)\sqrt{\hbar G/c^3} = 2M\sqrt{G^2/c^4} = 2GM/c^2$ (identity).

Appendix B Registry Checklist

Every number appearing in the text is checked one-to-one against the number registry (Section 2.4). Numbers not registered are marked as such together with their tier.

Table A1. Registry check of the numbers appearing in the text. "Dropped" means explicitly excluded by the registry filter; in that case the tier is stated in the right column.

No.	Where it appears	Registry status (bracket)	Tier
2	the 2 of $r_s = 2N\ell_p$ (Thm. 4)	a different quantity from the registered 2s (brackets, absent axes)	the cost-bearing step count fixed by Definition 5(a); distinguished from the registered 2 by Paper I filter rule 3
3	CAS 3 steps, $\ln 3$, 3π (Sec. 7)	registered (both lists)	DOF
4	$S = A/4$ (Thm. 1), ball value (Thm. 1, 2)	registered (in each list)	DOF; the separation of the two registrations is Theorem 1
5	five irreversible axes (Paper I)	registered (cost)	DOF
7	the 7 of $9 = 7 + 2$	registered (structural)	DOF
9	2^9 (Thm. 3)	registered (structural: $7 + 2$; cost: $13 - 4$)	DOF; used here on the structural side
13	total cost (Def. 1)	registered (cost)	DOF
128	2^7 (Paper I)	registered (structural)	DOF
512	the value of 2^9 (Thm. 3)	itself unregistered	binary state count of the registered exponent 9; counted by the exponent-registration rule (Sec. 5.3) - unlike $8 (= 2^3)$, it is the exponent, not the base, that is subject to registration
8	γ denominator, $\ln 3/(8\pi)$ (Sec. 7)	dropped (2^3 factorable)	the d-ring 8 bits of the data type $137 = 8 + 128 + 1$; confined to class III
10	$5120 = 10 \times 2^9$ (Thm. 3)	dropped (intermediate step)	$\dim SO(5)$, group-theoretic constant; class II
$\sqrt{3}$	γ numerator (Sec. 7)	dropped (norm derivative)	workbench three-axis norm (Axiom 2 proposition); confined to class III
π	throughout	outside the registry (geometric constant)	hemispheric arc (geometry of Axioms 2 and 3)

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Abbreviations

The following abbreviations are used in this manuscript:

CAS	conditional state transition (Compare-And-Swap)
R, C, S	observe (Read), compare (Compare), record (Swap)
QNM	quasinormal mode
LQG	loop quantum gravity
RLU	addressless virtual indexing unit
DOF	degrees of freedom

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